GI 11 · Intestinal microbiology

In-class station cases

Cases 1–6 · each question, then the answer with why every other option is wrong

Rotavirus | Norovirus | Cholera | B. cereus | Non-typhoidal Salmonella | STEC → HUS

Case 1

In January, a 2-month-old formula-fed infant has fever, repeated vomiting, and profuse diarrhea for 2 days.

The mother declares he has no known medical problems.

Physical examination reveals BT 39°C, BP 80/50 mmHg, PR 160/min, RR 45/min



Case 1 · Question 1

Which virus is the most likely causative agent?

A. Astrovirus

B. Norovirus

C. Rotavirus

D. Sapovirus

E. Enteric adenovirus

Case 1 · Answer 1

C. Rotavirus

Winter + infant + fever, vomiting first, profuse watery diarrhea with dehydration (fast pulse, sunken eyes) = rotavirus, the commonest cause of severe dehydrating diarrhea in young children. Formula-fed = no breast-milk secretory IgA.

Option	Why it's wrong
A. Astrovirus	Mild watery diarrhea in young children or the elderly; rarely this much fever or dehydration
B. Norovirus	All ages, outbreaks, vomiting > diarrhea; a single winter infant fits rotavirus better
D. Sapovirus	Same family as norovirus; milder illness and less common
E. Enteric adenovirus	Longer diarrhea (5–12 days), no season; this is 2 days and severe in winter

Case 1 · Question 2

Which of the following supports the diagnosis of this viral gastroenteritis?

A. Stool WBC 10 cells/HPF

B. Stool RBC 5 cells/HPF

C. Mucous stool

D. Stool pH 5.0

E. Negative stool reducing substance

Case 1 · Answer 2

D. Stool pH 5.0

Rotavirus destroys villus tips → lactase and other brush-border enzymes lost → undigested sugars reach the colon → bacteria ferment them to acids → stool pH below 5.5 and reducing substance 2+ or more.

Option	Why it's wrong
A. Stool WBC 10/HPF	White cells mean inflammatory (invasive bacterial) diarrhea; viral stool has none
B. Stool RBC 5/HPF	Blood means invasion or cytotoxin (Shigella, EHEC, Campylobacter)
C. Mucous stool	Mucus goes with colonic inflammation (dysentery)
E. Negative reducing substance	Opposite: undigested sugars make it positive

Case 1 · Question 3

What is the most appropriate laboratory investigation?

A. Electron microscopy

B. Viral culture

C. Antigen detection

D. Virus-specific IgM

E. RT-PCR

Case 1 · Answer 3

C. Antigen detection

Stool antigen (ELISA, latex agglutination, immunochromatography) is the test of choice for rotavirus: fast, cheap and sensitive, because children shed about 10^{11} particles per gram of stool.

Option	Why it's wrong
A. Electron microscopy	Shows the wheel-shaped virus, but slow, costly, reference labs only
B. Viral culture	Rotavirus grows poorly in routine cell culture
D. Virus-specific IgM	Serology is for population studies; diagnosis is from stool
E. RT-PCR	Very sensitive, but antigen is the routine first choice; RT-PCR is first choice for norovirus

Case 1 · Question 4

What is the most effective preventive measure against this viral gastroenteritis?

A. Inactivated vaccine

B. Live-attenuated vaccine

C. Breastfeeding

D. Hand washing

E. Wearing mask

Case 1 · Answer 4

B. Live-attenuated vaccine

All rotavirus vaccines are oral live-attenuated (Rotarix 2 doses; RotaTeq, Rotavac, Rotasiil 3 doses). They mimic natural infection, build gut secretory IgA and prevent most severe, dehydrating disease.

Option	Why it's wrong
A. Inactivated vaccine	None in use; protection needs gut IgA, not serum IgG
C. Breastfeeding	On the slide, but partial and short-lived protection
D. Hand washing	Helps, but rotavirus is very infectious and survives on surfaces
E. Wearing mask	Spread is fecal–oral, not airborne

Case 2

In December, a 25-year-old female develops fever, frequent vomiting, and diarrhea for 2 days. She declares no known medical problems. Two days ago, she ate seafood salad with her friends at a party. Ten of her friends also suffered from similar illness.

Physical examination reveals BT 38°C, BP 100/60 mmHg, PR 80/min, RR 20/min



Case 2 · Question 1

Which virus is the most likely causative agent?

A. Astrovirus

B. Norovirus

C. Rotavirus

D. Sapovirus

E. Enteric adenovirus

Case 2 · Answer 1

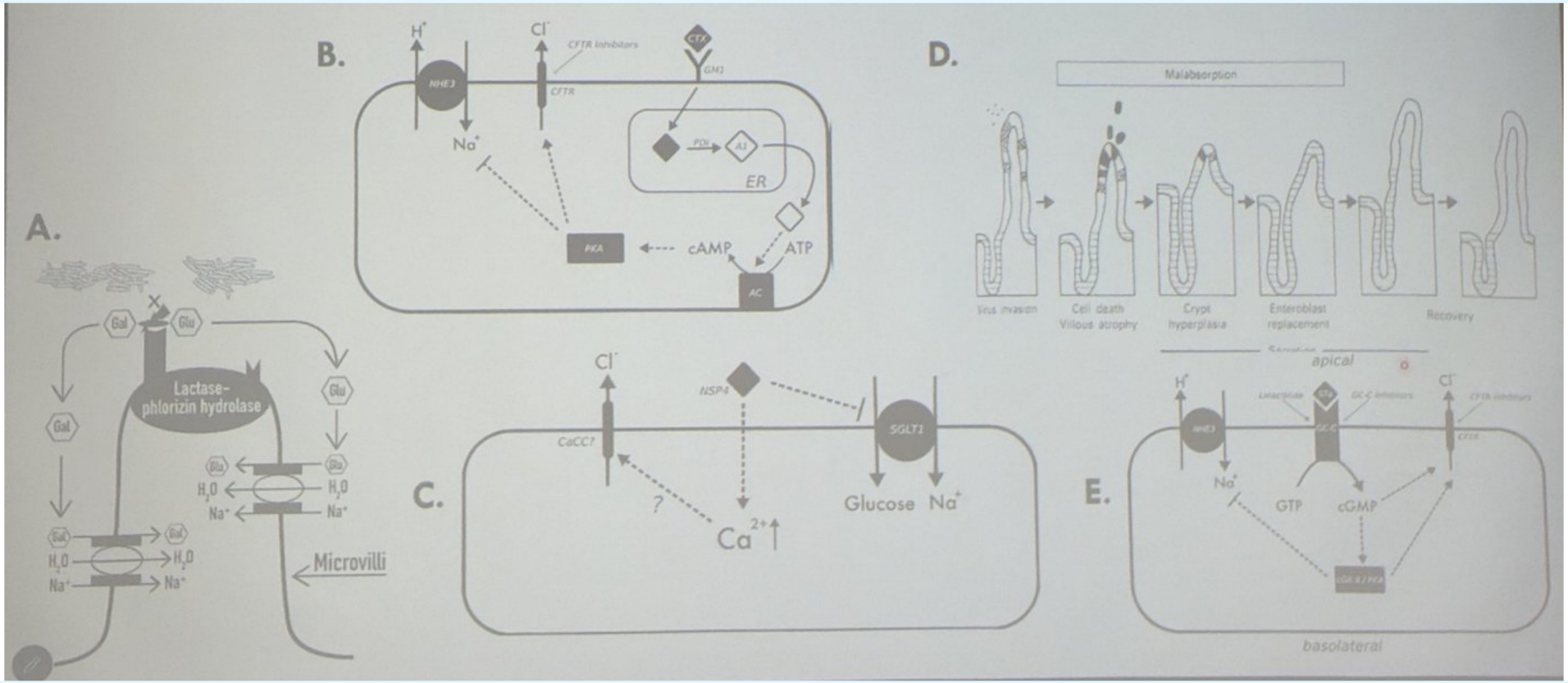
B. Norovirus

Adult + winter + outbreak after shared seafood salad + vomiting-predominant + onset within 1–2 days (incubation 10–50 h) = norovirus, the “winter vomiting disease”.

Option	Why it’s wrong
A. Astrovirus	Mild watery diarrhea in young children or the elderly
C. Rotavirus	Severe disease in infants and young children; adults rarely ill
D. Sapovirus	Same family as norovirus, but milder, mostly young children
E. Enteric adenovirus	Children under 2, no season, longer diarrhea (5–12 days)

Case 2 · Question 2

Which of the following is the most likely pathogenesis?



Case 2 · Answer 2

D. Villus damage → malabsorption

Norovirus infects villus-tip enterocytes → cells die, villi blunt → malabsorption → osmotic watery diarrhea. Crypt hyperplasia replaces the cells and the gut recovers in 1–3 days. No named toxin.

Picture	What it shows (why it's wrong)
A	Lactase loss → lactose malabsorption: a downstream effect, not the main mechanism
B	Cholera toxin: GM1 → cAMP → PKA → CFTR
C	Rotavirus NSP4: ↑Ca^{2+} → Cl^- secretion, SGLT1 blocked (the trap: rota has a toxin, noro doesn't)
E	ETEC heat-stable toxin: guanylate cyclase C → cGMP → CFTR

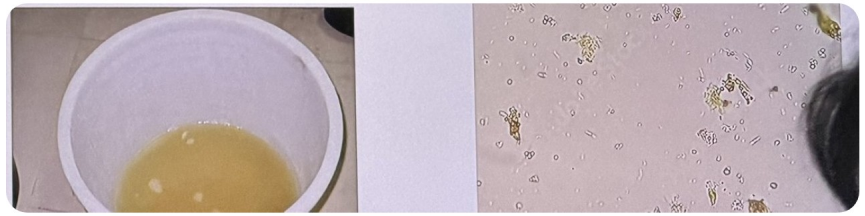
Case 3

A 10-year-old boy presents with a 2-day history of vomiting and profuse, rice-water diarrhea without abdominal pain.

BT 36°C, BP 60/40 mmHg, HR 120/min, RR 24/min

He is lethargic and has sunken eyes, a dry mouth, cold and clammy skin, and poor skin turgor.





Case 3 · Question 1

Which of the following is the most likely causative agent?

A. Enteropathogenic *E. coli*

B. *Campylobacter jejuni*

C. *Salmonella* Typhi

D. *Shigella dysenteriae*

E. *Vibrio cholerae*

Case 3 · Answer 1

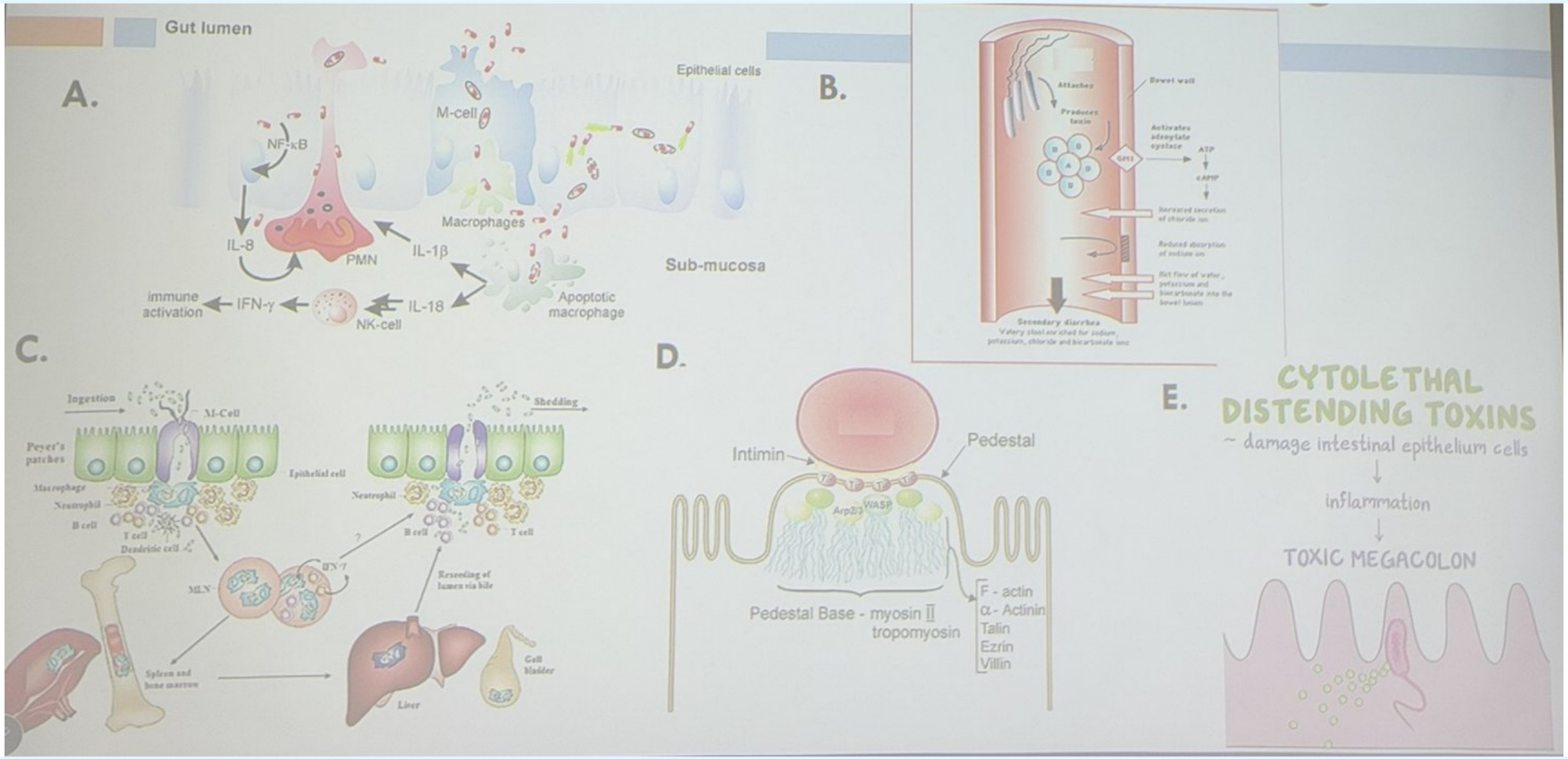
E. *Vibrio cholerae*

Profuse rice-water diarrhea + no fever (36°C) + no abdominal pain + hypovolemic shock = cholera. The toxin makes the gut secrete water without damaging cells, so no fever, pain or blood.

Option	Why it's wrong
A. EPEC	Watery diarrhea in infants under 2; not massive rice-water stool with shock
B. Campylobacter	Inflammatory, often bloody diarrhea with fever and abdominal pain
C. Salmonella Typhi	Enteric fever: high fever, abdominal pain, rose spots
D. Shigella dysenteriae	Dysentery: fever, cramps, tenesmus, small stools with blood and mucus

Case 3 · Question 2

Which of the following is the most likely pathogenesis?



Case 3 · Answer 2

B. Cholera toxin → ↑cAMP

B subunits bind GM1 → A subunit activates adenylate cyclase → ↑cAMP → CFTR opens → Cl^-, water, K^+ and HCO_3^- pour into the lumen, Na^+ absorption falls → rice-water diarrhea. No cell damage, so no fever or blood.

Picture	What it shows (why it's wrong)
A	Shigella: M cell entry, macrophage apoptosis, IL-1β/IL-8 → neutrophils → dysentery
C	Salmonella Typhi: Peyer's patches → lymph nodes → spleen, bone marrow, liver → gallbladder
D	EPEC/EHEC attaching and effacing: intimin and pedestal
E	Campylobacter cytolethal distending toxin → epithelial damage → inflammation

Case 3 · Question 3

What is the immediate management priority for this patient?

A. Administer antimotility drugs

B. Start antibiotics before fluid resuscitation

C. Administer oral rehydration solution and observe only

D. Begin immediate intravenous fluid resuscitation

E. Administer a proton pump inhibitor

Case 3 · Answer 3

D. Begin immediate IV fluid resuscitation

Severe dehydration with hypovolemic shock (BP 60/40, lethargic) → IV fluids first (e.g. Ringer's lactate)†, then oral rehydration solution once he can drink. Fluids save the life; antibiotics only help after.

Option	Why it's wrong
A. Antimotility drugs	Don't stop toxin-driven secretion; fluid loss is the threat
B. Antibiotics first	Antibiotics shorten illness and shedding, but only after fluids; shock kills first
C. ORS and observe	ORS is for mild to moderate dehydration; he is in shock
E. Proton pump inhibitor	No role; stomach acid actually protects against V. cholerae

Case 4

A 30-year-old woman develops abrupt nausea, repeated vomiting, and abdominal cramps 3 hours after eating fried rice that had been left at room temperature and reheated.

Onset: 3 hours

Main symptom: vomiting

Food: reheated fried rice

Case 4 · Question 1

Which of the following is the most likely causative agent?

A. Shiga toxin of STEC

B. Cholera toxin

C. Enterotoxigenic *E. coli*

D. Preformed heat-stable enterotoxin of *Bacillus cereus*

E. Heat-labile enterotoxin of *Staphylococcus aureus*

Case 4 · Answer 1

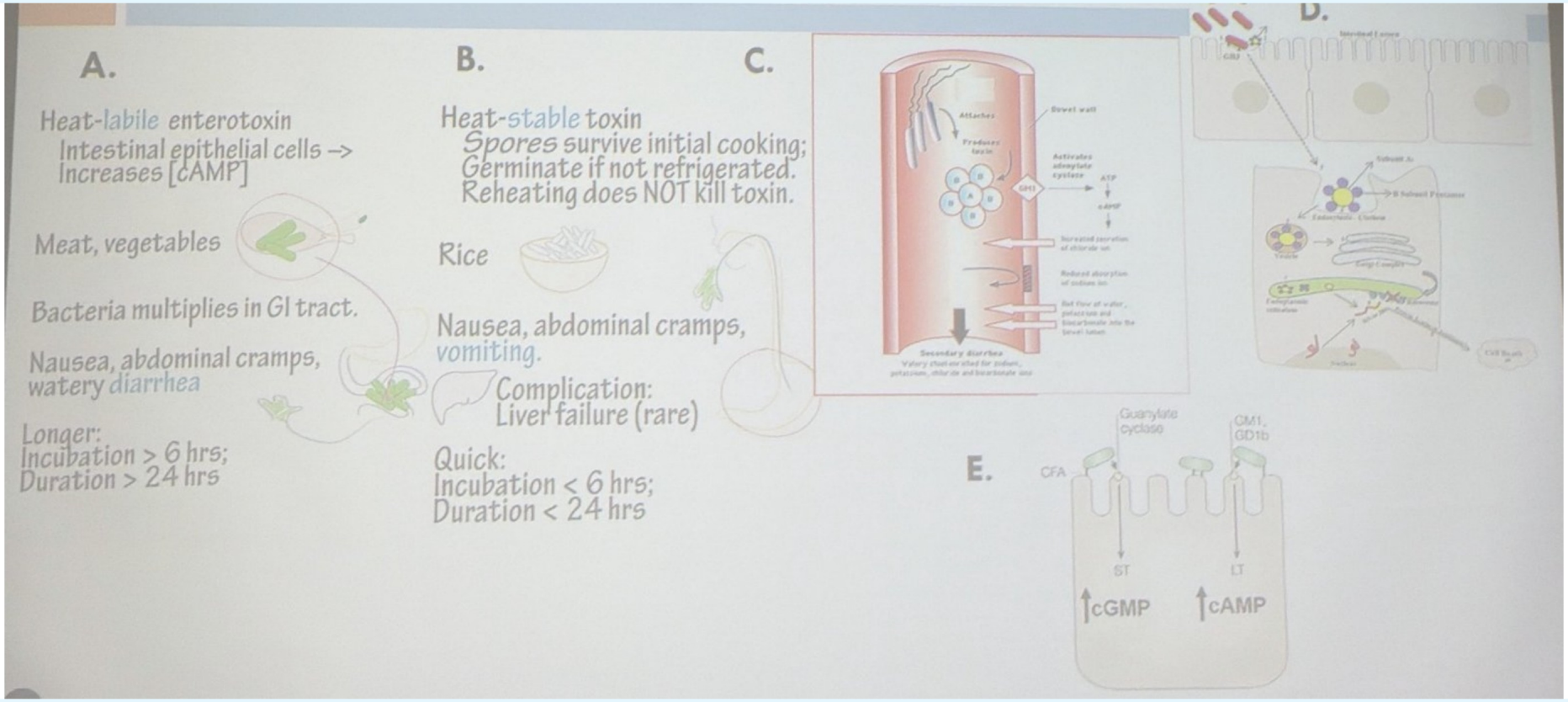
D. Preformed heat-stable enterotoxin of *B. cereus*

Vomiting 1–6 hours after reheated fried rice = emetic *B. cereus*. Spores survive cooking → germinate at room temperature → make cereulide in the rice → reheating does NOT destroy it.

Option	Why it's wrong
A. Shiga toxin of STEC	Bloody diarrhea after 3–4 days (undercooked beef), not vomiting at 3 hours
B. Cholera toxin	Rice-water diarrhea after more than 16 hours
C. ETEC	Watery traveler's diarrhea after more than 16 hours; bacteria must grow in the gut first
E. S. aureus heat-labile toxin	S. aureus toxin is heat-STABLE, and its foods are ham, mayonnaise, egg/potato salad, cream pastries

Case 4 · Question 2

Which of the following is the most likely pathogenesis?



Case 4 · Answer 2

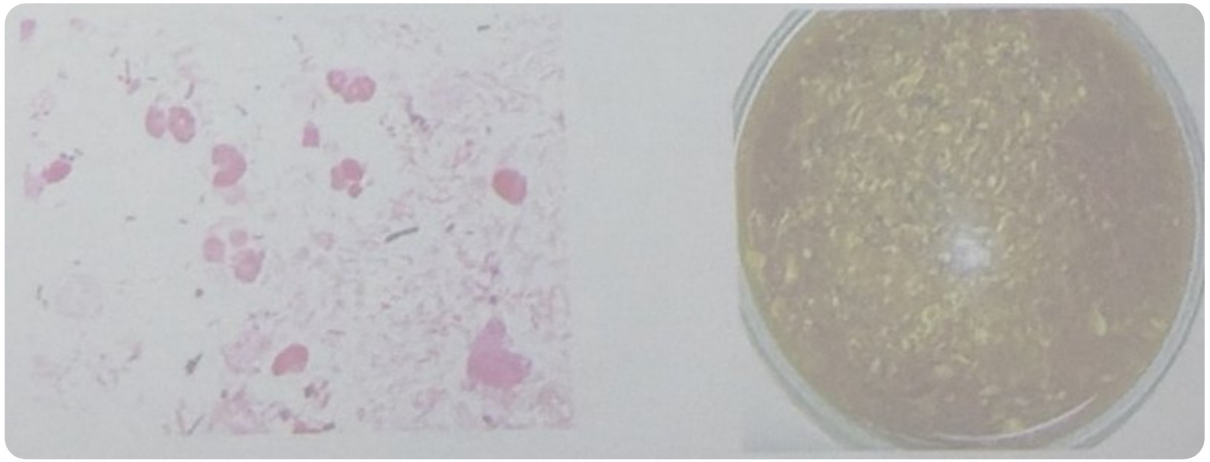
B. Heat-stable toxin (emetic form)

Cereulide is made in the rice before eating → irritates the gut → vagal afferents → vomiting centre → nausea and vomiting within 6 hours, over within 24 hours. Rare complication: liver failure.

Option	Why it's wrong
A. Heat-labile toxin	Diarrheal form: toxin made in the gut, 8–16 hours, watery diarrhea; meat and vegetables
C. Cholera toxin	GM1 → adenylate cyclase → ↑cAMP → secretory diarrhea
D. Shiga toxin	Binds Gb3 → stops ribosomes → cell death → HUS
E. ETEC ST / LT	ST → ↑cGMP, LT → ↑cAMP → watery traveler's diarrhea

Case 5

A 65-year-old man develops fever, abdominal cramps, nausea, and diarrhea with mucus 24 hours after eating undercooked eggs. Stool microscopy shows 50–100 neutrophils/HPF and no erythrocytes. He is alert and hemodynamically stable, has no history of vascular grafts or immunosuppression, and has not recently taken antibiotics.



Case 5 · Question 1

Which of the following is the most likely causative agent?

A. Enterotoxigenic *E. coli*

B. *Clostridioides difficile*

C. *Vibrio cholerae*

D. *Salmonella* Typhi

E. Non-typhoidal *Salmonella*

Case 5 · Answer 1

E. Non-typhoidal *Salmonella*

Undercooked eggs + fever, cramps and mucus diarrhea 24 hours later + 50–100 neutrophils/HPF (inflammatory diarrhea) = NTS gastroenteritis. M cells → IL-8 → neutrophils; usually contained locally.

Option	Why it's wrong
A. ETEC	Toxin-mediated watery diarrhea with no stool neutrophils
B. C. difficile	Needs recent antibiotics or a hospital stay; he has had no antibiotics
C. V. cholerae	Non-inflammatory rice-water diarrhea, no fever, no neutrophils
D. Salmonella Typhi	Enteric fever after 7–14 days (systemic), not gastroenteritis at 24 hours

Case 5 · Question 2

What is the appropriate treatment?

A. Activated charcoal

B. Antibiotic administration

C. Antidiarrheal drug

D. Fluid and electrolyte replacement

E. Proton pump inhibitor

Case 5 · Answer 2

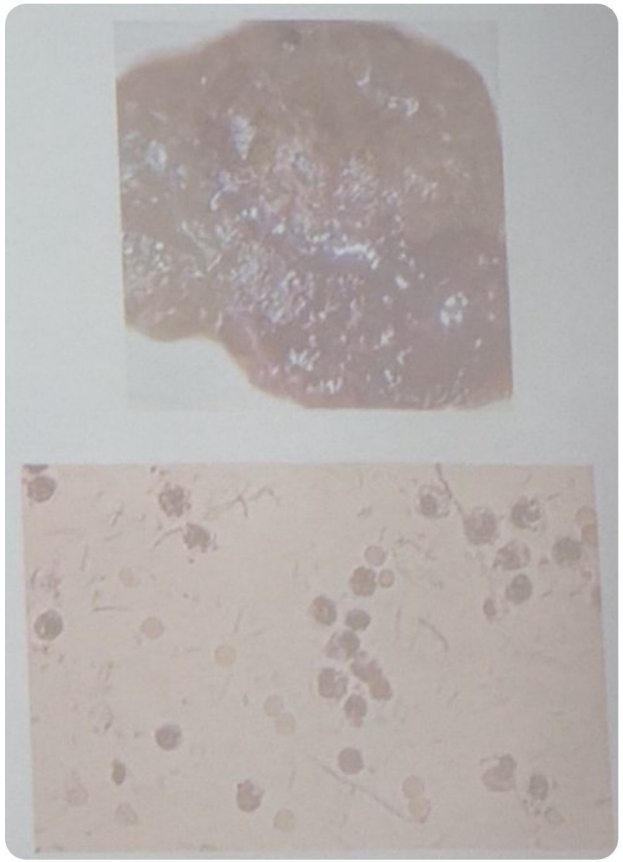
D. Fluid and electrolyte replacement

Uncomplicated, non-invasive NTS → fluids and electrolytes; it settles by itself. Complicated or invasive disease (bacteremia, infants, elderly with vascular grafts, immunosuppressed) → antibiotics.

Option	Why it's wrong
A. Activated charcoal	For some poisonings and overdoses, not bacterial gastroenteritis
B. Antibiotics	Not needed when uncomplicated; can prolong carriage. He has no high-risk features
C. Antidiarrheal drug	Avoided in inflammatory diarrhea with fever; keeps invasive bacteria in the gut
E. Proton pump inhibitor	No role; low stomach acid raises the risk of Salmonella

Case 6

A 15-year-old girl has 3 days of bloody diarrhea and abdominal pain after eating a hamburger. She receives an antibiotic. One week after diarrhea began, she develops pallor and reduced urine output. Tests show microangiopathic hemolytic anemia, thrombocytopenia, and elevated creatinine.



Case 6 · Question 1

What is the most likely diagnosis?

A. Acute glomerulonephritis

B. Post-infectious nephritis

C. Hemolytic Uremic Syndrome (HUS)

D. Idiopathic thrombocytopenic purpura (ITP)

E. Thrombotic thrombocytopenic purpura (TTP)

Case 6 · Answer 1

C. Hemolytic uremic syndrome (HUS)

Bloody diarrhea after a hamburger, then about a week later the triad: microangiopathic hemolytic anemia + thrombocytopenia + acute kidney injury (high creatinine, low urine).

Option	Why it's wrong
A. Acute glomerulonephritis	Hematuria, edema, hypertension; no hemolytic anemia or low platelets
B. Post-infectious nephritis	Follows a throat or skin infection; no microangiopathic anemia or thrombocytopenia
D. ITP	Low platelets only; no hemolysis, no kidney injury
E. TTP	Also anemia + low platelets, but adults with fever and neurological signs, no preceding bloody diarrhea†

Case 6 · Question 2

What is the most likely causative agent?

A. *Streptococcus pyogenes*

B. Shiga toxin-producing *Escherichia coli* (STEC)

C. *Shigella flexneri*

D. *Clostridioides difficile*

E. *Campylobacter jejuni*

Case 6 · Answer 2

B. Shiga toxin-producing *E. coli* (STEC)

Undercooked ground beef (hamburger) + bloody diarrhea → HUS = STEC (EHEC O157:H7). Lab clue: sorbitol-negative on sorbitol MacConkey.

Option	Why it's wrong
A. S. pyogenes	Causes post-streptococcal glomerulonephritis, not bloody diarrhea after beef
C. Shigella flexneri	Dysentery, but no Shiga toxin (only S. dysenteriae type 1 makes it)
D. C. difficile	Follows antibiotics → pseudomembranous colitis; here the antibiotic came after the diarrhea
E. Campylobacter	Bloody diarrhea from poultry → Guillain–Barré, not HUS

Case 6 · Question 3

What is the most likely pathophysiological mechanism?

A. Immune-complex deposition in glomeruli

B. Shiga toxin-mediated renal microvascular endothelial injury

C. Direct bacterial invasion of the kidney

D. Autoantibodies against red blood cells

E. Viral infection of renal epithelial cells

Case 6 · Answer 3

B. Shiga toxin-mediated renal microvascular endothelial injury

Toxin absorbed → binds Gb3 on glomerular endothelium → A subunit cleaves 28S rRNA → protein synthesis stops → endothelial injury → microthrombi → red cells sheared (schistocytes), platelets used up, kidney injury.

Option	Why it's wrong
A. Immune complexes	Mechanism of post-infectious glomerulonephritis
C. Bacterial invasion	STEC stays in the gut; only the absorbed toxin reaches the kidney
D. RBC autoantibodies	Autoimmune hemolytic anemia (Coombs positive), not microangiopathic
E. Viral infection	No virus; this is a toxin complication of bacterial diarrhea

Case 6 · Question 4

What is the mechanism that best explains the worsening of his symptoms?

A. Selection of an antibiotic-resistant STEC strain

B. Direct antibiotic destruction of red blood cells

C. Direct antibiotic toxicity to glomerular cells

D. Increased expression or release of Shiga toxin

E. Immune-complex deposition in the kidney

Case 6 · Answer 4

D. Increased expression or release of Shiga toxin

Antibiotics stress STEC → SOS response → the phage carrying the Shiga toxin genes switches on → more toxin made; dying bacteria burst and release stored toxin† → more toxin reaches the kidney → worse HUS. So no antibiotics in STEC.

Option	Why it's wrong
A. Resistant strain	Resistance means the drug fails; it doesn't make HUS worse
B. Antibiotic kills RBCs	Red cells are sheared in damaged small vessels, not destroyed by the drug
C. Antibiotic kidney toxicity	Kidney injury is toxin-mediated endothelial damage
E. Immune complexes	Post-infectious glomerulonephritis, not HUS

Summary: clue → organism → key point

Case	Key clues	Organism	Must remember
1	Winter, infant, fever, vomiting, watery stool, dehydration	Rotavirus	Stool pH < 5.5; stool antigen; live oral vaccine
2	Adult, winter, outbreak after seafood salad, vomiting	Norovirus	Villus damage, no toxin; RT-PCR
3	Rice-water stool, no fever, no pain, shock	Vibrio cholerae	Toxin → ↑cAMP; IV fluids first
4	Vomiting 3 h after reheated fried rice	B. cereus (emetic)	Preformed heat-stable cereulide
5	Eggs, 24 h, fever, mucus, many stool neutrophils	Non-typhoidal Salmonella	Fluids; antibiotics only if invasive or high risk
6	Hamburger, bloody diarrhea, then anemia, low platelets, high creatinine	STEC → HUS	Shiga toxin → Gb3 endothelium; antibiotics make it worse

Incubation tells you where the toxin is made: 1–6 h = preformed in food (*S. aureus*, *B. cereus* emetic) · 8–16 h = made in the gut (*C. perfringens*, *B. cereus* diarrheal) · over 16 h = bacteria growing in the gut